

Investment Strategy Tool

Architecture and Technical Documentation

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1 Introduction

This document outlines the architecture, logic, and operational guidelines of a custom investment and portfolio management tool. The underlying strategy is designed for a specific investor profile and operates under a defined set of market and personal premises. The fundamental assumptions driving the development and execution of this strategy are as follows:

- **Systematic Monthly Contributions:** The strategy assumes the investor makes regular, monthly capital injections. This allows the system to leverage dollar-cost averaging and continuously allocate new capital to optimize portfolio weights without forcing the sale of existing assets.
- **Zero-Cost Data Infrastructure:** At present, the investor is not willing to incur recurring subscription costs for premium financial data providers. Consequently, the architecture is strictly constrained to rely on free, publicly available sources (such as Yahoo Finance and the Kenneth French Data Library). The system compensates for the inherent unreliability and rate limits of free APIs through robust internal data cleaning, dynamic local caching, and fallback mechanisms.
- **Low Liquidity Constraints:** The invested capital is designated for long-term growth. The investor does not require immediate or short-term access to liquidity, eliminating the need to maintain a significant cash drag and preventing forced liquidations during market downturns.
- **Sensitivity to Operational Frictions:** The tool explicitly incorporates the negative impact of operational frictions, specifically taxes and brokerage commissions. Consequently, the algorithm penalizes excessive asset rotation, favoring a buy-and-hold approach or utilizing new monthly contributions for rebalancing rather than triggering taxable events through frequent selling.
- **Solid Diversified Baseline:** The investor's total wealth is not solely parked in risk-free assets. A foundational layer of the portfolio is already allocated to globally diversified Exchange-Traded Funds (ETFs). This baseline provides a stable, low-volatility return profile across the broader market.
- **Controlled Volatility Exposure for Alpha:** Building upon the stable global ETF foundation, the primary objective of this specific tool is to actively select and manage assets that introduce a slightly higher, calculated level of market volatility. The ultimate goal is to capture a premium over standard global ETFs, marginally beating the market while keeping the overall risk profile strictly contained.

2 Algorithmic Architecture and Execution Pipeline

The investment strategy is executed through a strictly defined, six-step computational pipeline. This pipeline runs periodically (e.g., monthly) to re-evaluate the investment universe, rebalance the portfolio, and deploy new capital optimally.

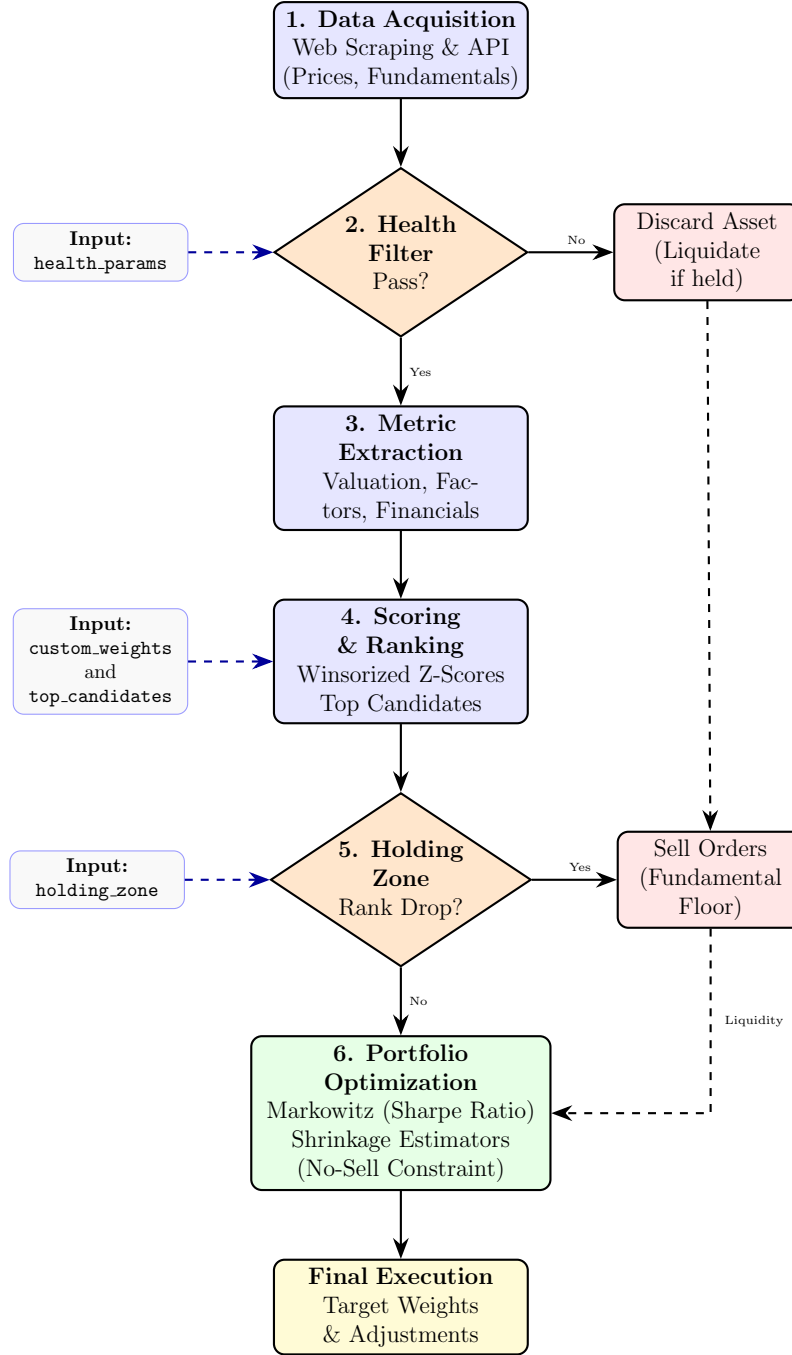
1. **Data Acquisition and Universe Definition:** The process begins by dynamically web-scraping the current constituents of major equity indices: the S&P 500, STOXX Europe 600, and IBEX 35. Once the raw ticker lists are consolidated, the system interfaces with financial APIs to download historical price action, fundamental accounting data (balance sheets, income statements, cash flows), and market factor returns.
2. **Structural Financial Health Filter:** Before any valuation takes place, the expanded universe is passed through a discrete, pass/fail financial health filter to eliminate companies with severe structural risks. The passing criteria are governed by customizable hyperparameters. The metrics evaluated include:
 - Positive Return on Assets (ROA)
 - Positive Operating Income (EBIT)
 - Improving ROA (Year-over-Year Trajectory)
 - Positive Revenue Growth (Year-over-Year)
 - Decreasing Leverage (Year-over-Year Debt-to-Equity reduction)
 - Improving Short-Term Liquidity (Current Ratio Year-over-Year)
 - Safe Interest Coverage Ratio
3. **Metric Extraction and Calculation:** For the assets that survive the health filter, the system calculates a comprehensive suite of continuous variables, categorized into three main pillars:
 - *Fundamental Valuation Multipliers:* EV/EBITDA, EV/EBIT, Free Cash Flow (FCF) Yield, Earnings Yield, EV/UFCE (Unlevered Free Cash Flow), EV/NOPAT, EV/Invested Capital (IC), and EV/(EBITDA - CAPEX).
 - *Continuous Financial Metrics:* ROA, ROA Trajectory, Revenue Growth, Debt-to-Equity, Debt-to-Equity Trajectory, Current Ratio, Current Ratio Trajectory, and Interest Coverage.
 - *Market Performance and Factor Metrics:* Fama-French 6-Factor Alphas (measuring persistent structural advantage), Factor Betas (Market, Size, Value, Profitability, Investment, and Momentum), and standard trailing momentum (e.g., 12M-1M).
4. **Cross-Sectional Scoring and Candidate Selection:** The extracted metrics are standardized using a Winsorized Z-score relative to their specific sectors to ensure sector-neutral evaluations. These Z-scores are aggregated into a 'Total Master Score' using a predefined set of hyperparameter weights. The universe is then ranked, and the highest-scoring assets are selected as the *Top Candidates* for the current period.
5. **Liquidation Protocol (Fundamental Floor):** The existing portfolio is cross-referenced against the newly evaluated universe. Sell orders are strictly triggered by two conditions:
 - An asset fails the discrete Structural Financial Health Filter (Step 2).

- An asset’s overall percentile rank drops below the predefined *Holding Zone* hyperparameter (e.g., dropping below the 80th or 90th percentile).

Capital from liquidated assets is collected and added to the user’s monthly fiat contribution.

6. **Cash-First Bounded Optimization:** With the final list of active holdings and new *Top Candidates* established, the system computes the optimal capital allocation using Modern Portfolio Theory (Markowitz). To handle estimation errors, the algorithm utilizes Shrinkage estimators: the Ledoit-Wolf model for the covariance matrix and the James-Stein (or Jorion Bayes-Stein) estimator for expected returns.

The optimizer maximizes the Sharpe Ratio with a strict *No-Sell Lower Bound Constraint*. The minimum weight for any currently held asset is mathematically locked to its current diluted market value over the new total portfolio value. This forces the optimizer to achieve the highest possible risk-adjusted return by strictly allocating the available liquidity (new contributions plus liquidated cash) without triggering unwanted taxable sales of healthy assets.



3 Hyperparameter Calibration and Backtesting Engine

To ensure the strategy is mathematically sound and capable of navigating real-world constraints, the algorithm relies on a robust historical backtesting engine. This engine acts as the objective function for hyperparameter calibration, simulating trade-by-trade operations over a sliding historical window (typically the last 12 months). The optimizer searches for the parameter combination that maximizes the final **monthly mean Net Return on Investment (Net ROI)**. The mean Net ROI is used to pursue a stable growth each month.

3.1 Operational Frictions and Tax-Aware Evaluation

Unlike theoretical models that assume frictionless markets, this backtesting engine strictly incorporates capital degradation caused by active management:

- **Brokerage Fees:** A fixed operational fee (e.g., 1 EUR per executed trade for Trade Republic) is deducted for every buy and sell order. Fractional share dustings are filtered out to prevent artificial trade spamming.
- **FIFO Tax Lot Tracking:** The system tracks individual share purchases as distinct tax lots (First-In, First-Out).
- **Capital Gains Tax (Spanish Brackets):** When a liquidation occurs (e.g., an asset falls through the fundamental floor), the algorithm calculates the exact realized gain or loss against the corresponding tax lot. Profits are subjected to the Spanish capital gains tiers: 19% up to 6,000 EUR, 21% up to 50,000 EUR, and 23% for amounts exceeding 50,000 EUR.

By factoring in these costs, the optimizer naturally penalizes hyper-active strategies, forcing it to find a balance between holding strong assets and rotating into better opportunities only when the expected alpha covers the fiscal drag.

3.2 Calibration Space: Tunable Hyperparameters

The quantitative strategy relies on multiple parameters that govern its strictness, ranking priorities, and structural behavior. The following parameters are exposed to the optimization algorithm:

3.2.1 Z-Score Ranking Weights (`custom_weights`)

These weights dictate the importance of each metric in the cross-sectional scoring matrix. The optimizer fine-tunes the balance between:

- **Value & Multipliers:** EV/UFCF, FCF Yield, EV/EBIT, Earnings Yield, and EV/EBITDA-CAPEX.
- **Financial Quality:** ROA, Interest Coverage, Debt-to-Equity, and their respective year-over-year trajectories.
- **Market Factors:** Fama-French 6-Factor Alpha and Market Beta.

3.2.2 Selection and Retention Constraints

These parameters dictate portfolio concentration and turnover frequency:

- **top_candidates:** The exact number of top-ranked equities the portfolio is allowed to hold (e.g., 5 to 10 assets).
- **percentile_exit_threshold (Holding Zone):** The minimum acceptable percentile rank an existing holding can drop to before a liquidation is triggered. A strict threshold (e.g., 90th percentile) increases turnover, while a relaxed threshold (e.g., 70th percentile) mimics a buy-and-hold approach.

3.2.3 Health Filter Thresholds (health_params)

These parameters define the baseline survival requirements before an asset can be scored:

- **roa_min & ebit_min:** The minimum acceptable Return on Assets and Operating Income (usually anchored at 0.0 to eliminate cash-burning companies).
- **interest_coverage_min:** The minimum ratio to ensure the company can service its debt without stress (e.g., 1.5x).
- **pass_threshold_standard:** The discrete number of health signals (out of 7) a standard company must pass to avoid immediate liquidation.

3.2.4 Temporal Lookback Windows

The amount of historical data injected into the estimators impacts model responsiveness:

- **FUNDAMENTAL_WINDOW_YEARS:** The number of years of accounting data used to compute structural trajectories (e.g., 3 years).
- **MARKOWITZ_WINDOW_YEARS:** The lookback period used to compute expected returns and the Ledoit-Wolf covariance matrix for the Sharpe Ratio optimization (e.g., 1 year).

4 Quantitative Metrics and Mathematical Definitions

This section details the mathematical formulations and the directional logic for all the continuous variables extracted by the algorithm. The cross-sectional scoring system normalizes these metrics to rank the investment universe.

4.1 Base Components

Several valuation multipliers rely on the concept of Enterprise Value (EV) and other derived structural components. They are defined as follows:

- **Enterprise Value (EV):** Represents the total theoretical takeover value of the business.

$$EV = MarketCapitalization + TotalDebt - CashandEquivalents$$

- **Unlevered Free Cash Flow (UFCF):** Cash generated by the core business operations, stripped of leverage effects.

$$UFCF = OCF - CAPEX + (|InterestExpense| \times (1 - TaxRate))$$

- **Net Operating Profit After Taxes (NOPAT):** Unlevered profitability.

$$NOPAT = EBIT \times (1 - TaxRate)$$

- **Invested Capital (IC):** Total capital deployed in the business operations.

$$IC = TotalDebt + Stockholders'Equity - CashandEquivalents$$

4.2 Fundamental Valuation Multipliers

These metrics assess whether an asset is fundamentally cheap or expensive relative to the cash or profit it generates.

- **EV / EBITDA**

- *Formula:* $EV \div EBITDA$
- *Description:* A standard proxy for cash-flow valuation, ignoring capital structure and non-cash depreciation/amortization.
- *Logic:* **Lower is better.** Indicates the company is undervalued relative to its core earnings.

- **EV / EBIT**

- *Formula:* $EV \div EBIT$
- *Description:* Similar to EV/EBITDA but accounts for the real cost of capital expenditure depreciation.
- *Logic:* **Lower is better.**

- **Free Cash Flow (FCF) Yield**

- *Formula:* $(OCF - CAPEX) \div MarketCapitalization$
- *Description:* Measures the pure cash return a theoretical investor would receive if all free cash flow were distributed as dividends.
- *Logic:* **Higher is better.**

- **Earnings Yield**

- *Formula:* $EBIT \div EV$
- *Description:* The inverse of the EV/EBIT ratio; expresses valuation as a percentage yield, allowing comparison against bond yields or risk-free rates.
- *Logic:* **Higher is better.**

- **EV / UFCF**

- *Formula:* $EV \div UFCF$
- *Description:* A strict valuation metric assessing the cost of the entire enterprise relative to the pure cash it generates for all stakeholders (debt and equity).
- *Logic:* **Lower is better.**

- **EV / NOPAT**

- *Formula:* $EV \div NOPAT$
- *Description:* Valuation against standardized post-tax operating profits.
- *Logic:* **Lower is better.**

- **EV / Invested Capital (EV/IC)**

- *Formula:* $EV \div IC$
- *Description:* Evaluates market premium relative to the actual capital deployed by the business.
- *Logic:* **Lower is better.**

- **EV / (EBITDA - CAPEX)**

- *Formula:* $EV \div (EBITDA - CAPEX)$
- *Description:* A practical cash-flow multiple that aggressively penalizes companies with high maintenance capital expenditure requirements.
- *Logic:* **Lower is better.**

4.3 Structural Financial Health Metrics

These metrics evaluate operational efficiency, liquidity, and balance sheet safety.

- **Return on Assets (ROA) & ROA Trajectory**

- *Formula:* $NetIncome \div TotalAssets$
- *Trajectory:* $ROA_{CurrentYear} - ROA_{PreviousYear}$
- *Description:* Measures the efficiency of asset utilization to generate profit. The trajectory rewards companies with improving operational leverage.
- *Logic:* **Higher is better.**

- **Debt-to-Equity (D/E) & D/E Trajectory**

- *Formula:* $TotalDebt \div Stockholders'Equity$
- *Trajectory:* $D/E_{CurrentYear} - D/E_{PreviousYear}$
- *Description:* Assesses structural balance sheet risk. The trajectory rewards corporate deleveraging (debt reduction).
- *Logic:* **Lower is better** (a negative trajectory value indicates debt reduction, adding positive points to the Z-score).

- **Current Ratio & Current Ratio Trajectory**

- *Formula:* $CurrentAssets \div CurrentLiabilities$
- *Trajectory:* $CR_{CurrentYear} - CR_{PreviousYear}$
- *Description:* A primary indicator of short-term liquidity and the ability to cover imminent obligations.
- *Logic:* **Higher is better.**

- **Interest Coverage Ratio**

- *Formula:* $EBIT \div |InterestExpense|$
- *Description:* A pure survival metric indicating how many times the operating profit can pay the debt service costs.

- *Logic:* **Higher is better.**

- **Revenue Growth**

- *Formula:* $(TotalRevenue_{Current} \div TotalRevenue_{Previous}) - 1$
- *Description:* Year-over-year top-line expansion rate.
- *Logic:* **Higher is better.**

4.4 Market Performance and Factor Metrics

These variables assess the asset’s historical behavior against the broader market and known academic risk premia.

- **Fama-French 6-Factor Alpha (α_{6F})**

- *Extraction:* Annualized intercept from an Ordinary Least Squares (OLS) regression against the Fama-French 5 Factors (Market, Size, Value, Profitability, Investment) plus Momentum.
- *Description:* Represents the excess return of the asset that cannot be explained by standard market risk exposures. It is a proxy for persistent structural advantage or idiosyncratic quality.
- *Logic:* **Higher is better.**

- **Market Beta (β_{Mkt})**

- *Extraction:* The coefficient of the *Mkt-RF* factor from the OLS regression.
- *Description:* Measures the asset’s volatility relative to the broader market index.
- *Logic:* **Lower is better.** The algorithm shows a slight preference for lower volatility to stabilize the downstream Ledoit-Wolf covariance matrix and Sharpe Ratio optimization.

- **Momentum (12M - 1M)**

- *Formula:* Risk-adjusted composite of 12-month and 6-month log returns, excluding the most recent month to avoid mean reversion.
- *Description:* Captures the medium-term price trend strength relative to its annualized volatility.
- *Logic:* **Higher is better.**